EduTrack – System Design Document

1. Introduction

1.1 Purpose

The purpose of this System Design Document (SDD) is to describe the architecture and design of the **EduTrack system**, including system components, database structure, modules, and interactions between system elements.

This document serves as a reference for developers and stakeholders involved in building and maintaining the system.

1.2 System Overview

EduTrack is a **web-based academic management system** designed to manage and track student academic information such as attendance, assignments, courses, and marks.

The system provides a centralized platform for both **students and teachers** to manage academic activities efficiently.

Key features include:

- Student academic record management
- Course management
- Attendance tracking
- Assignment management
- Marks management

The system uses modern web technologies to provide a scalable and efficient academic management solution.

1.3 Technology Stack

Layer	Technology
Frontend	React.js
Backend	Spring Boot
Database	MySQL
API Type	REST APIs
Version Control	Git & GitHub

2. System Architecture (High Level Design)

EduTrack follows a **three-tier architecture**, separating the application into three main layers.

2.1 Presentation Layer

Technology: **React.js**

Responsibilities:

- Provides user interface
- Handles user input
- Displays academic data
- Sends API requests to backend

Users interact with the system through this layer.

2.2 Application Layer

Technology: **Spring Boot**

Responsibilities:

- Handles business logic
- Processes requests from frontend
- Performs authentication and authorization
- Communicates with database
- Exposes REST APIs

2.3 Data Layer

Technology: **MySQL**

Responsibilities:

- Stores system data
- Maintains relationships between entities
- Ensures data consistency

2.4 High Level Architecture Diagram

```
Users (Students / Teachers)
        |
        v
Frontend (React.js)
        |
        v
Backend (Spring Boot REST APIs)
        |
        v
Database (MySQL)
```

Users interact with the system through the frontend interface. The frontend communicates with the backend through REST APIs, and the backend retrieves or stores data in the database.

2.5 Sequence Diagram – User Login and Data Retrieval

```
Student        Frontend        Backend API        Database
   |               |                |                 |
1. Login Request   |                |                 |
   |-------------->|                |                 |
   |               | 2. Send Login Credentials       |
   |               |--------------->|                 |
   |               |                | 3. Verify User  |
   |               |                |--------------->|
   |               |                |                 |
   |               |                | 4. Return User Data
   |               |                |<---------------|
   |               | 5. Login Success Response       |
   |               |<---------------|                 |
   | 6. View Dashboard              |                 |
   |-------------->|                |                 |
   |               | 7. Request Academic Data        |
   |               |--------------->|                 |
   |               |                | 8. Fetch Data   |
   |               |                |--------------->|
   |               |                |                 |
   |               |                | 9. Return Data  |
   |               |                |<---------------|
   |               | 10. Send Data to UI             |
   |               |<---------------|                 |
   | 11. View Attendance / Marks / Assignments      |
```

The sequence diagram illustrates the interaction between the **student, frontend interface, backend API, and database** during the login and academic data retrieval process.

3. Detailed System Design (Low Level Design)

3.1 System Modules

The EduTrack system consists of the following modules:

Student Module

Functions:

- Student login
- View enrolled courses
- View attendance
- View assignments
- View marks

Teacher Module

Functions:

- Teacher login
- Manage courses
- Upload assignments
- Mark attendance
- Enter student marks

Course Management Module

Functions:

- Add new courses
- Assign teachers
- Manage course information

Attendance Management Module

Functions:

- Record student attendance
- View attendance reports

Assignment Management Module

Functions:

- Create assignments
- Manage assignment deadlines
- View assignments

Marks Management Module

Functions:

- Enter student marks
- Update marks
- View performance results

3.2 Database Design

The system database contains the following entities:

Students

Attributes:

- student_id (Primary Key)
- name
- email
- password
- department
- year

Teachers

Attributes:

- teacher_id (Primary Key)
- name
- email
- password
- department

Courses

Attributes:

- course_id (Primary Key)
- course_name
- credits
- teacher_id (Foreign Key)

Relationship:
One teacher can teach multiple courses.

Attendance

Attributes:

- attendance_id (Primary Key)

- student_id (Foreign Key)
- course_id (Foreign Key)
- date
- status

Assignments

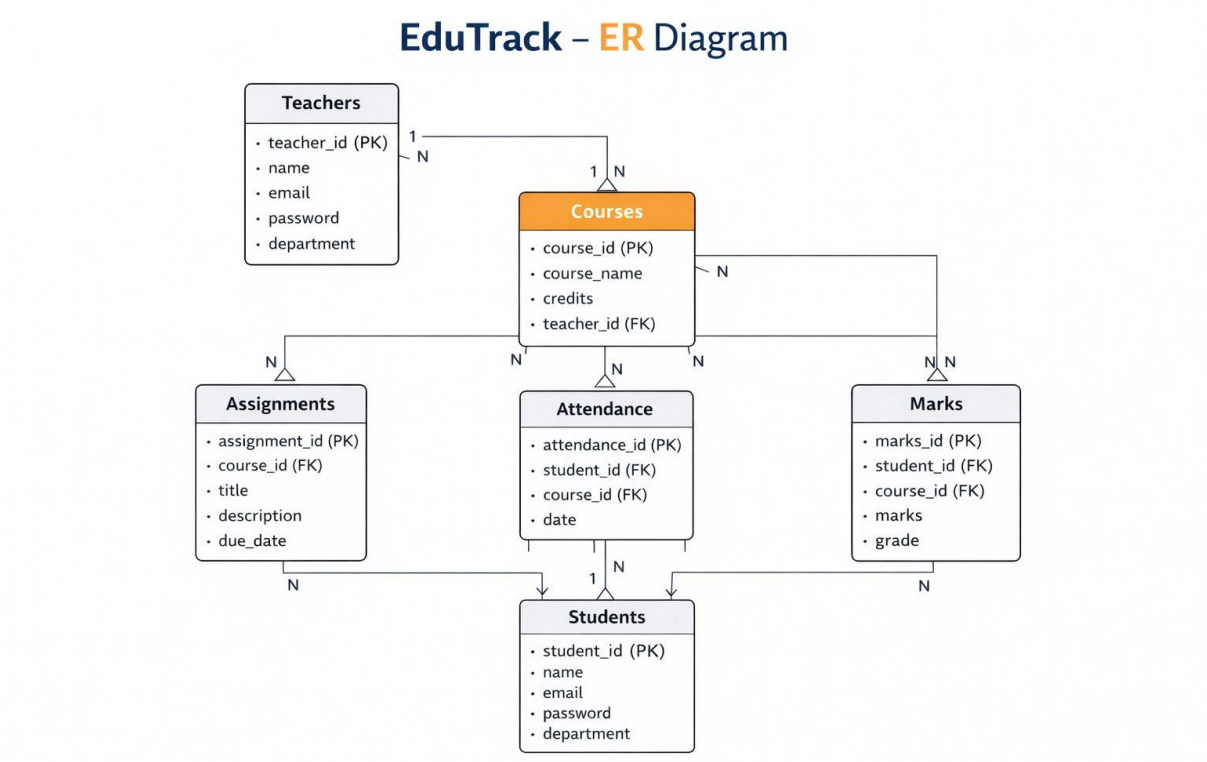
Attributes:

- assignment_id (Primary Key)
- course_id (Foreign Key)
- title
- description
- due_date

Marks

Attributes:

- marks_id (Primary Key)
- student_id (Foreign Key)
- course_id (Foreign Key)
- marks
- grade



4. API Design

The backend provides REST APIs used by the frontend.

Example APIs:

API	Method	Description
/api/auth/login	POST	User login
/api/students	GET	Get student details
/api/courses	GET	Get courses
/api/attendance	POST	Mark attendance
/api/assignments	POST	Create assignment
/api/marks	POST	Add student marks

5. Non-Functional Requirements

Performance

The system should provide fast response times and handle multiple users efficiently.

Security

Security mechanisms include:

- Secure authentication
- Password encryption
- Role-based access control

Scalability

The system should support an increasing number of users without performance degradation.

Reliability

The system should ensure consistent operation with minimal downtime.

Usability

The interface should be simple and user-friendly.

Maintainability

The system should be easy to maintain and allow developers to add new features without affecting existing modules.

6. Conclusion

EduTrack provides an efficient solution for managing academic records and tracking student performance. By integrating technologies such as React.js, Spring Boot, and MySQL, the system offers a scalable and reliable academic management platform.

The system automates tasks such as attendance tracking, assignment management, and marks recording, improving efficiency and transparency in academic institutions.